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Lab 3 : Particle Swarm Optimization (PSO)

Summary : Particle Swarm Optimization (PSO) algorithm is inspired by bird flocking or fish schooling & used to optimize mathematical function by iteratively improving the current value of function, used to optimize math functions on basis of pre defined fitness qualities

Pseudo Code / Algorithm

1) Initialize : the Math function to be optimised

2) Initialize parameters : The parameters to determine function like start, inertial speed etc

3) Initialize population : Generate a sample population for starting with random points

4) Evaluate fitness : The current points value within function is evaluated

5) Update velocity & position : On the basis of the qualities initialised / determine previously update current value with global and local maxima

6) Iterate : The current function is iterated again for either certain number of times or until convergence is met

7) Output : Return the best value among all iterations

formula : $P_i = ((P_{Best} + distance) + (G_{Best} + distance)) * inertia$

Neural Network

Summary of paper:

Paper IEEE : • Maximizing English Teaching Efficiency with Particle Swarm optimization - driven neural network training

• proposed by Wu, F., & Chen, Y. (2004)

Summary:

. The paper by Fei Wu & Yu Chen, proposes the integration of PSO with Neural Network to improve effectiveness of English language teaching, the approach uses PSO to optimise training of nn by searching for suitable weight parameters based on students learning data.

• A learning capability model is used to analyze students learning characteristics & abilities, after which neural network is trained on PSO approach and achieve low error

Objective:

• The main objective of paper is to develop the & evaluate PSO based neural network training method for improving the effectiveness of english language teaching

• The study aims to overcome limitations of conventional teaching and learning assessment methods by using NN to analyze students learning characteristics and to optimize the training process

• The trained model analyzes students learning pattern and allows teachers to help student improve language

Code:

```
Initialise training data
def sigmoid(x):
    return 1/(1+ exp(-x))
def neural-network (weight, input):
    seperate hidden weights, biases & output and
    Calculate hidden layer
    return predicted output
def fitness (weights):
    return mean error as fitness
def initialize_particles()
    generate random position & velocities
    set each particles personal best. return swarm
def update_particle_velocity (particle):
    Calculate updated velocity
    Add updated velocity to current position
    new position is new neural weights
def pso():
    return best weights and minimised errors
```

Output

```
Iteration : 0  : 0.23336
Iteration 10 :  0.18427
Iteration 20:  0.170445
Iteration 30 :  0.1619
Iteration 40 :  0.155711
Iteration 50 :  0.1645771
```

final result

Best error: 0.155711

proceed
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